DRUGS USED IN EPILEPSY

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Epilepsy

- A chronic disorder characterized by repeated/recurrent attacks of abnormal electrical discharge of cerebral neurons resulting in EEG, sensory, motor, autonomic and physiological changes;

- Each attack is called a **seizure** or a **fit**;

- Epilepsy affects 0.5 - 1% of population, second most common neurologic disorder after stroke

Causes of Epilepsy

I. Primary Epilepsy: - No known cause (idiopathic)

II. Secondary Epilepsy:

- Local causes: - **Head trauma, Meningitis, Tumors in the brain;**
- Systemic causes: - **Hypoglycemia, hypocalcemia, fevers**
- Drug-induced: - e.g. **Insulin, Tricyclic Antidepressants, CNS stimulants and Antipsychotics, etc**

Pathophysiology of Epilepsy

Hyper-excitability of cerebral neurons in brain region manifesting an epileptic foci may be due to:

1. Change in brain neurotransmitters or their signal transduction
2. Increased activity of excitatory brain neurotransmitters *(e.g. Glutamate)*
3. Decreased activity of inhibitory brain neurotransmitters *(e.g. GABA)*
4. Increased membrane permeability to ions *(e.g.* Ca^{2+}, Na^{+}*)*.

Classification of Epilepsy

Partial (Focal) Seizures

1. **Simple Partial Epilepsy**: - *localized motor (convulsions) or sensory disturbances.*

2. **Complex Partial *(Psychomotor)* Epilepsy**:- *attacks of confused behavior with disturbances of consciousness and hallucinations.*

Generalized Seizures

1. **Tonic-clonic (Grand-mal) Epilepsy:** - *convulsions involving the whole body with loss of consciousness.*

2. **Absence Seizures (Petit-mal Epilepsy):** - *brief abrupt loss of consciousness (10-15 sec) with blinking of eyelids or staring in children).*

3. **Myoclonic Epilepsy:** - *Sudden brief single or repetitive muscle contractions without loss of consciousness.*

Mode of action of Anti-epileptic drugs

Anti-epileptic drugs decrease hyper excitability of cerebral neurons by one of the following mechanisms:

1) Change in brain transmitters:

- Facilitation of GABA action *(e.g.* ***Phenobarbital, Benzodiazepines)****;*
- Increasing GABA levels by decreasing its breakdown through inhibition of transaminase enzyme *(e.g.* ***Valproate and Vigabatrin)****;*

2) Decrease neuronal membrane ion permeability:

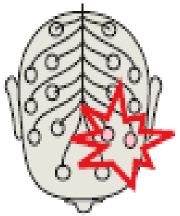
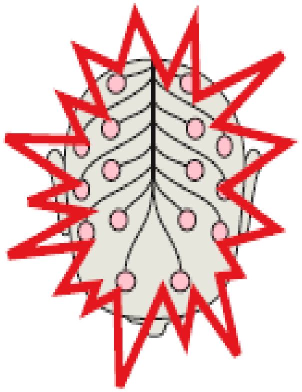
- Block Na^+ channels *(e.g.* ***Phenytoin, Carbamazepine and Valproate)****;*
- Block Ca^{2+} channels in the thalamus *(e.g* ***Ethosuximide).***

- **STATUS EPILETICUS: -** Continuous successions of acute non-stop seizures (>20 minutes) without any period of recovery.

- **It is a medical emergency**

- *Drug of choice is Diazepam IV while Phenobarbital IV can be given as follow up*

Indications for Anti-epileptic drugs

		I.	II.	III. Choice
Focal seizures 	Simple seizures	Carbam-azepine	Valproic acid, Phenytoin	Primidone, Phenobarbital
	Complex or secondarily generalized	+ Lamotrigine or Vigabatrin or Gabapentin	+ Lamotrigine or Vigabatrin or Gabapentin	+ Lamotrigine or Vigabatrin or Gabapentin
Generalized attacks 	Tonic-clonic attack (grand mal) Tonic attack Clonic attack Myoclonic attack	Valproic acid	Carbamazepine, Phenytoin	Lamotrigine, Primidone, Phenobarbital
		+ Lamotrigine or Vigabatrin or Gabapentin	+ Lamotrigine or Vigabatrin or Gabapentin	+ Lamotrigine or Vigabatrin or Gabapentin
	Absence seizure	Valproic acid	Ethosuximide	
			+ Lamotrigine or Clonazepam	+ Lamotrigine or Clonazepam

Luellmann, 2005

Drug Name	Clinical Use	Adverse Effects
Lamotrigine	Partial seizures	*Insomnia, somnolence, ataxia, dizziness, diplopia, etc*
Carbamazepine	Tonic-clonic seizures, Mixed seizures, Psychomotor seizures; Neuropathic pain	*Aplastic anemia, blood dyscrasias, dizziness, drowsiness, nausea & vomiting, diplopia, hepatotoxicity*
Ethosuximide	Absence seizures	*Hematologic disturbances, anorexia, urticaria, drowsiness, ataxia, vomiting, etc*
Valproate (*Valproic acid*)	Tonic-clonic; Absence seizures	*Nausea, vomiting, rash, sedation, indigestion, nystagmus, diplopia, etc*
Phenytoin	Partial seizures, Tonic-clonic seizures	*CNS depression, hypotension, rash, nystagmus, ataxia, gingival hyperplasia, blood dyscrasias, etc*

Other Anti-epileptic drugs

- Trimethadione
- Clorazepate
- Felbamate
- Gabapentin
- Magnesium sulfate
- Oxcarbazepine
- Primidone
- Zonisamide

Phenytoin and phosphenytoin

- Like carbamazepine, these drugs block voltage-sensitive sodium channels by prolonging the inactivation state of these channels
- These drugs exhibit dose dependent kinetics where lower doses are eliminated by first order kinetics while higher concentrations saturate biotransformation and exhibit zero order kinetics

Adverse effects

- Phenytoin causes a number of adverse effects like megaloblastic anaemia due to its interference with folate metabolism
- Gingival hopoplasia where the gums extend down over the teeth
- Steven johnsons syndrome and toxic epidermal necrosis
- Oteomalacia due to the interfering with vitamin D metabolism
- Diplopia, slurred speech due to inter interference with cerebellar function

Phenytoin polymorphism and need for serum therapeutic monitoring

- Phenytoin hydroxylation exhibits polymorphism, this factor being responsible for the considerable variation in the plasma concentration produced by a given dose

- Because of these effects and variations, serum drug levels should be monitored at the start of therapy and wherever toxicity and therapeutic failure occurs

Principles of drug therapy

- Attempts should be made in most cases to control seizures with a single drug therapy(Monotherapy)
- This helps to minimize side effects, reduce costs and increase patient compliance
- Unless the patient is experiencing frequent seizures, it is usually best to start drug and to start with a low dose which should be gradually increased until therapeutic dose has been reached or the occurring of the intolerable adverse effect
- If a single drug has significantly reduced the occurrence but has not eliminated them, it is prudent to add another drug to the regimen than switching to a new drug
- Two drugs acting by different mechanism may control seizures when a single drug is not adequate

Contraindications & Interactions

- Known hypersensitivity
- Additive CNS depression with other CNS depressant drugs *(e.g. alcohol, narcotic analgesics, and antidepressants)*
- Phenytoin contraindicated in pregnancy; in patients with sinus bradycardia, SA node and AV node block, respectively
- Carbamazepine and Succinimides contraindicated in patients with bone marrow depression or hepatic or renal impairment and during lactation

END